

PPO (Proximal Policy Optimization) — quick reference

1. Objective: $L^{\text{CLIP}} = E[\min(r_t A_t, \text{clip}(r_t, 1-\text{eps}, 1+\text{eps}) A_t)]$
2. Probability ratio: $r_t(\theta) = \pi_{\theta}(a_t|s_t) / \pi_{\theta_{\text{old}}}(a_t|s_t)$
3. Typical eps in [0.1, 0.3]
4. Use GAE for advantage A_t with lambda in [0.92, 0.98]
5. Use clipped value loss to stabilise critic updates
6. Mini-batch size and number of epochs per update are sensitive

Used in rotpen-rdd v3 with eps=0.2, lambda=0.95.